

Digital Egypt Pioneers Initiative

Round 4
2025 - 2026

رواد مصر الرقمية

الدورة الرابعة
2025 - 2026

Track: AI & Data Science - Microsoft Machine Learning Engineer

Group Code: ONL4_AIS2_S7

Instructor: Eman Elgalad

Company Name: Skills Dynamix

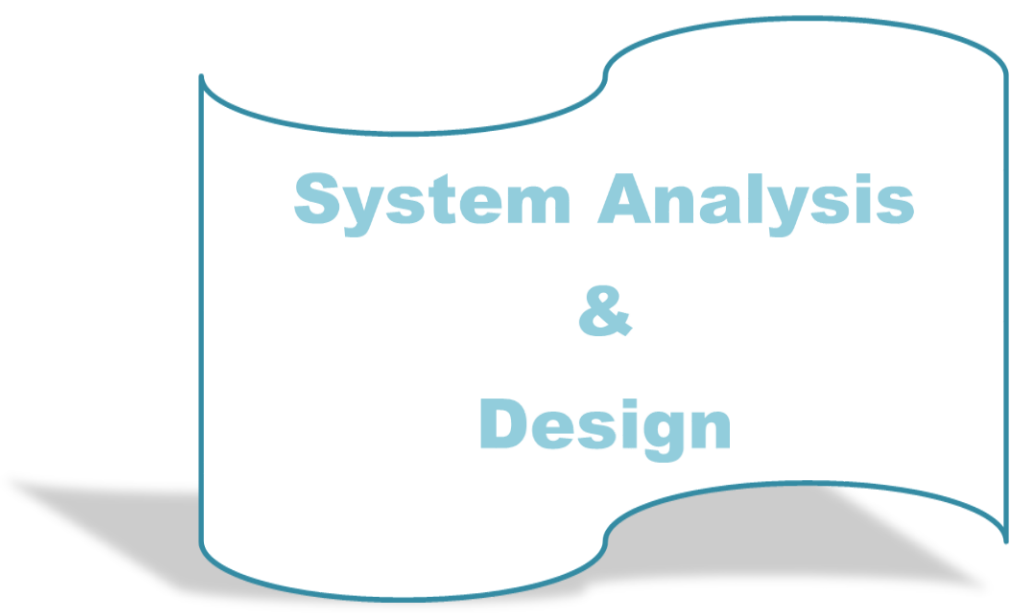


Project Name

"AI-Powered Predictive Maintenance for Industrial Equipment"

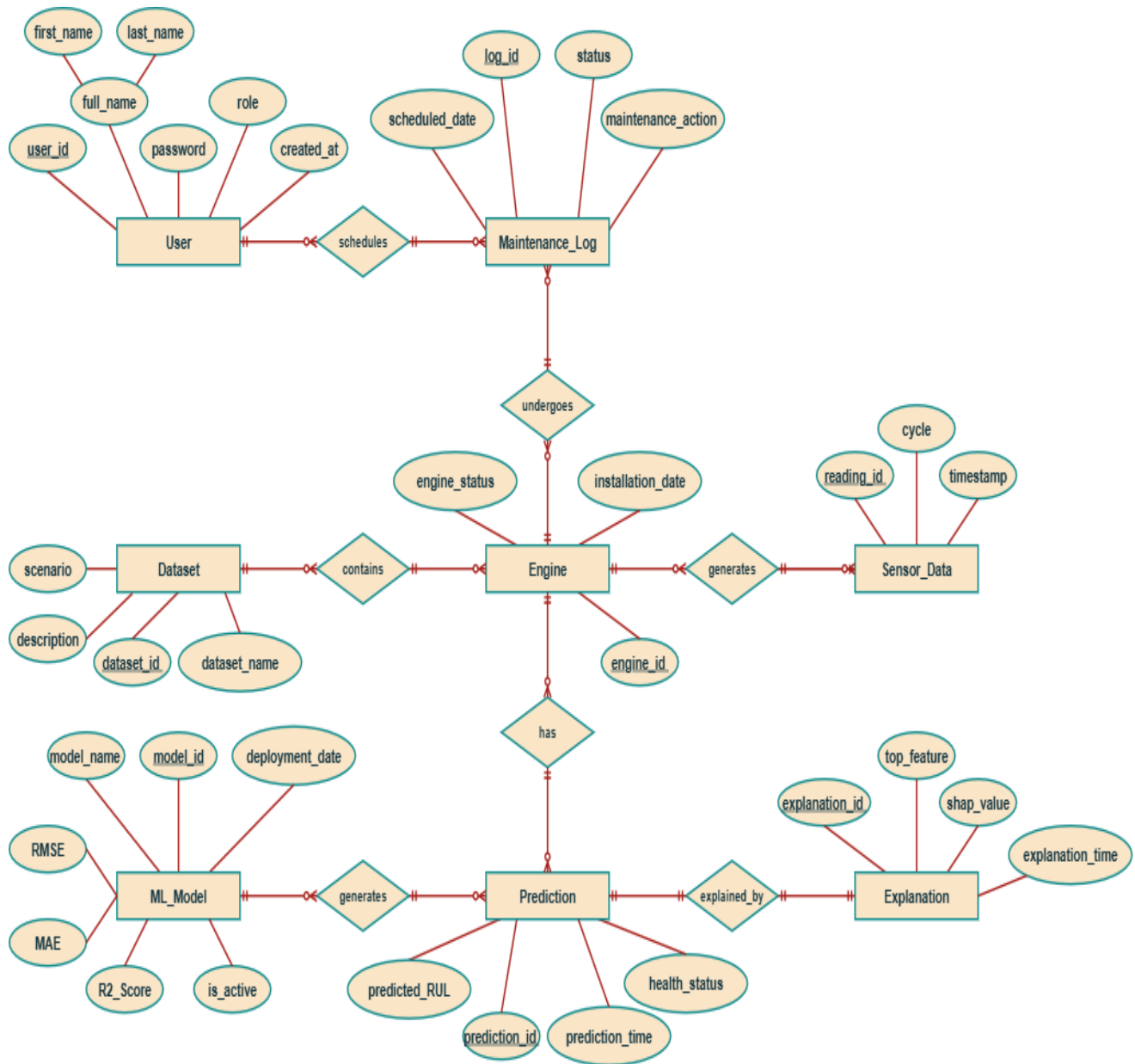
No.	Name
1	Moaz Osama Esamil Hataba (TL)
2	Abrar Ashraf Shaban Ahmad
3	Marwan Muhammad Darwish Ammar
4	Yussef Ahmad Muhammad Elshamy
5	Salma Muhammad Nageh Abdul-Hamid
6	Nour-Eldin Kamel Muhammad Hasanein

Task		Page No.	Evaluation
Entity-Relationship Diagram		5	
Logical Schema		6	
Physical Schema		7	
Normalization		9	

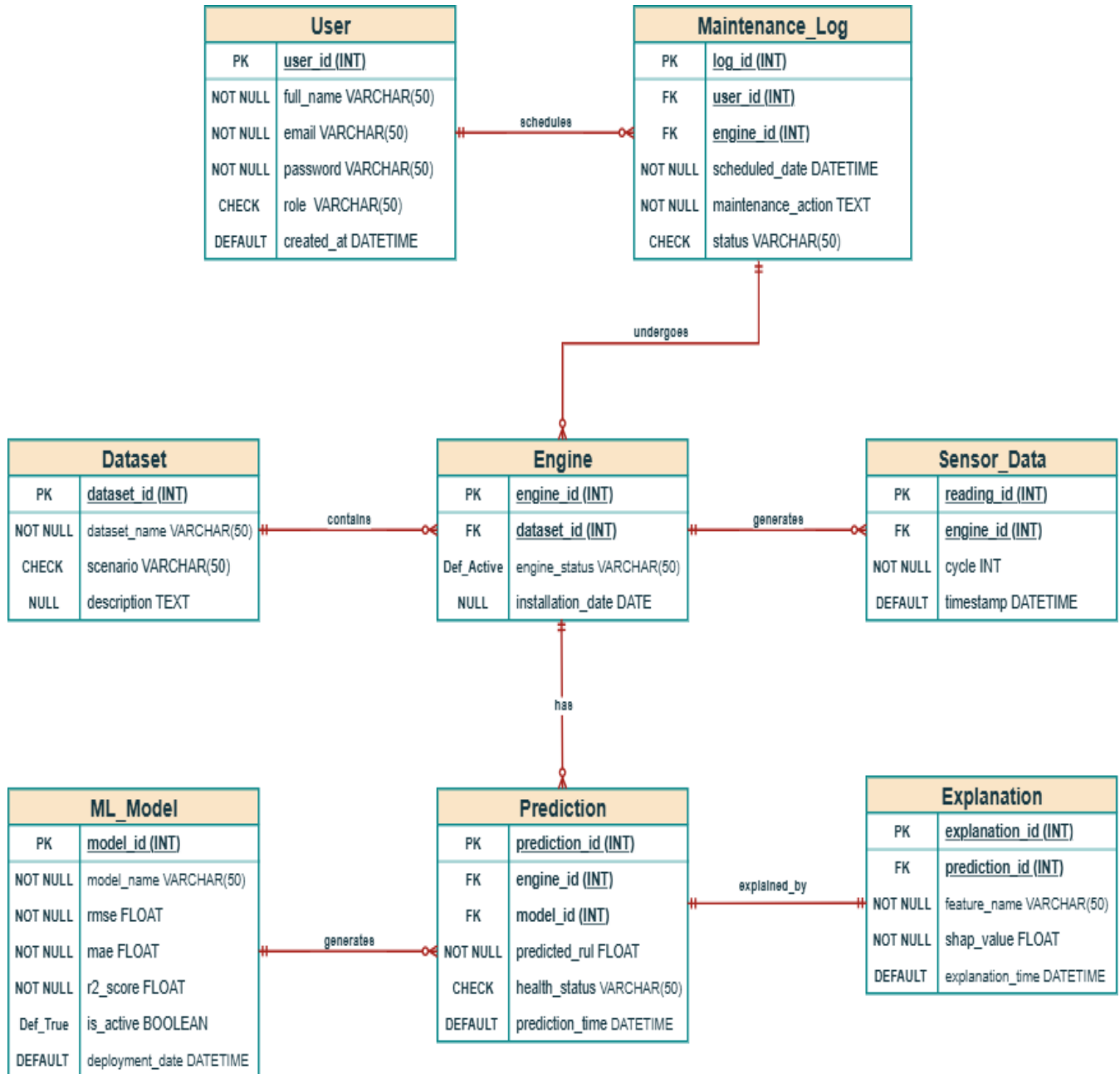


System Analysis & Design

Entity-Relationship Diagram (ERD):



Logical Schema:



Physical Schema:

- **USER TABLE**

```
CREATE TABLE User (  
    user_id INT PRIMARY KEY AUTO_INCREMENT,  
    full_name VARCHAR(100),  
    email VARCHAR(100) UNIQUE NOT NULL,  
    password VARCHAR(255) NOT NULL,  
    role VARCHAR(50),  
    created_at DATETIME DEFAULT CURRENT_TIMESTAMP  
);
```

- **MAINTENANCE LOG TABLE**

```
CREATE TABLE Maintenance_Log (  
    log_id INT PRIMARY KEY AUTO_INCREMENT,  
    user_id INT,  
    engine_id INT,  
    scheduled_date DATE,  
    maintenance_action TEXT,  
    status VARCHAR(50),  
    FOREIGN KEY (user_id) REFERENCES User(user_id),  
    FOREIGN KEY (engine_id) REFERENCES Engine(engine_id)  
);
```

- **DATASET TABLE**

```
CREATE TABLE Dataset (  
    dataset_id INT PRIMARY KEY AUTO_INCREMENT,  
    dataset_name VARCHAR(100),  
    scenario VARCHAR(10),  
    description TEXT  
);
```

- **ENGINE TABLE**

```
CREATE TABLE Engine (  
    engine_id INT PRIMARY KEY AUTO_INCREMENT,  
    dataset_id INT,  
    engine_status VARCHAR(50),  
    installation_date DATE,  
    FOREIGN KEY (dataset_id) REFERENCES Dataset(dataset_id)  
);
```

- **SENSOR DATA TABLE**

```
CREATE TABLE Sensor_Data (  
    reading_id INT PRIMARY KEY AUTO_INCREMENT,  
    engine_id INT,  
    cycle INT,  
    timestamp DATETIME,  
    FOREIGN KEY (engine_id) REFERENCES Engine(engine_id)  
);
```

- **ML MODEL TABLE**

```
CREATE TABLE ML_Model (  
    model_id INT PRIMARY KEY AUTO_INCREMENT,  
    model_name VARCHAR(100),  
    RMSE FLOAT,  
    MAE FLOAT,  
    R2_Score FLOAT,  
    deployment_date DATE,  
    is_active BOOLEAN  
);
```

- **PREDICTION TABLE**

```
CREATE TABLE Prediction (  
    prediction_id INT PRIMARY KEY AUTO_INCREMENT,  
    engine_id INT,  
    model_id INT,  
    predicted_RUL FLOAT,  
    health_status VARCHAR(50),  
    prediction_time DATETIME,  
    FOREIGN KEY (engine_id) REFERENCES Engine(engine_id),  
    FOREIGN KEY (model_id) REFERENCES ML_Model(model_id)  
);
```

- **EXPLANATION TABLE**

```
CREATE TABLE Explanation (  
    explanation_id INT PRIMARY KEY AUTO_INCREMENT,  
    prediction_id INT,  
    top_feature VARCHAR(100),  
    shap_value FLOAT,  
    explanation_time DATETIME,  
    FOREIGN KEY (prediction_id) REFERENCES Prediction(prediction_id)  
);
```


Normalization:

- **Unnormalized Form (UNF)**

- Initially, all information could be stored in a single table.
- This table contains:
 - UserID.
 - UserName.
 - Role.
 - Dataset.
 - EngineID.
 - Cycle.
 - Sensor1 – 21.
 - PredictedRUL.
 - HealthStatus.
 - SHAPFeature.
 - SHAPValue.
 - ModelName.
 - RMSE.
 - MAE.
 - R².
 - MaintenanceAction.
- Problems
 - Data redundancy.
 - Update anomalies.
 - Insert anomalies.
 - Delete anomalies.
 - Multiple unrelated entities stored together.

UserID	UserName	Role	Dataset	EngineID	Cycle	Sensor1 – 21	PredictedRUL
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HealthStatus	SHAPFeature	SHAPValue	ModelName	RMSE	MAE	R ²	MaintenanceAction
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● First Normal Form (1NF)

- Rule:
 - Must already satisfy 1NF.
 - Remove partial dependencies.
- Improvement
 - Each sensor value is atomic.
 - Each reading has a unique primary key.

ReadingID (PK)	EngineID	Cycle
1	5	1
2	5	2

● Second Normal Form (2NF)

- Rule:
 - Atomic values only
 - No repeating groups
 - Every row uniquely identified.

- User Table

UserID (PK)	Name	Email	Role
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- Maintenance_Log Table

LogID (PK)	EngineID (FK)	UserID (FK)	Action	Status
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- Dataset Table

DatasetID (PK)	Scenario	Description
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- Engine Table

EngineID (PK)	DatasetID (FK)	Status
---------------	----------------	--------

- Sensor_Data Table

ReadingID (PK)	EngineID (FK)	Cycle	Sensor1 - 21
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- ML_Model Table

ModelID (PK)	ModelName	RMSE	MAE	R ²
--------------	-----------	------	-----	----------------

- Prediction Table

PredictionID (PK)	EngineID (FK)	ModelID (FK)	PredictedRUL	HealthStatus
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- Explanation Table

ExplanationID (PK)	PredictionID (FK)	FeatureName	SHAPValue
--------------------	-------------------	-------------	-----------

● Third Normal Form (3NF)

- Rule:
 - Must satisfy 2NF.
 - Remove transitive dependencies.
- User Table

user_id (PK)	full_name	email	password	role	created_at
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- Maintenance_Log Table

log_id (PK)	engine_id (FK)	user_id (FK)	status	maintenance_action	scheduled_date
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- Dataset Table

dataset_id (PK)	dataset_name	Scenario	Description
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- Engine Table

engine_id (PK)	dataset_id (FK)	Status	installation_date
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- Sensor_Data Table

reading_id (PK)	engine_id (FK)	cycle	sensor1 - 21
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- ML_Model Table

model_id (PK)	model_name	rmse	mae	r2_score	deployment_date	is_active
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- Prediction Table

prediction_id (PK)	engine_id (FK)	model_id (FK)	predicted_rul	health_status	prediction_time
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- Explanation Table

explanation_id (PK)	prediction_id (FK)	feature_name	shap_value	explanation_time
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